AIRWISE User Manual

AIRWISE: Environmental Decision-Support & Response System *Version 2.0 Document Date: October 11, 2026*

Section 2 — Project Introduction

AIRWISE is a campus-focused environmental decision-support system designed to help educational institutions and organizations in Delhi-NCR make operational decisions based on air quality conditions.

The primary problem AIRWISE addresses is the gap between abstract air quality data and actionable, accountable institutional response. By combining real-time environmental data with an auditable decision engine, AIRWISE allows Campus Administrators and Coordinators to quickly determine if outdoor activities are safe, select mitigation scenarios, and execute verifiable response plans.

AIRWISE establishes operational accountability. It does not establish clinical medical advice, nor does it guarantee absolute safety from environmental hazards. It provides decision support using best-available modelled forecasts and optional ground measurements.

Section 3 — Key Concepts and Terminology

- **Campus**: A predefined physical location (e.g., Ghaziabad, Noida) being monitored for environmental conditions.
- **Pollutants (PM2.5, PM10, NO2, O3, SO2, CO)**: Particulate matter and gases measured in the air. PM2.5 and PM10 are primary drivers of air quality concern.
- **Modelled forecast versus direct observation**: Forecasts (provided by Open-Meteo) are computational predictions of conditions across a grid cell. Direct observations (e.g., via OpenAQ) are actual readings from physical sensors. Modelled forecasts are not exact point measurements.
- **Campus assessment**: An evaluation of the current air quality data against predefined thresholds for a specific activity, producing a risk status (e.g., "Proceed", "Review Required", "Action Recommended").
- **Evidence and data freshness**: The exact snapshot of pollutant levels at the time of the assessment. Freshness indicates how recently the data was updated.
- **Data Readiness Score**: A metric reflecting the completeness and freshness of the underlying data. It is not a calibrated probability of model accuracy.
- **Recommendation and scenario**: Suggested actions (e.g., Move Indoors, Reschedule) presented by the Decision Simulator based on the assessment's risk level.
- **Response plan**: A structured set of tasks generated by choosing a scenario.
- **Decision receipt and verification**: A cryptographically hashed record (using SHA-256) of the assessment evidence and chosen alternative. It is tamper-evident, allowing users to verify that a plan's underlying environmental justification has not been altered post-creation.
- **Administrative approval**: The step where a Campus Administrator authorizes a draft Response Plan for execution.
- **Response tasks and task status**: Individual actionable items (Pending, In Progress, Completed) assigned to roles like Facilities Coordinator or Sports Coordinator.
- **Operational outcome versus measured environmental impact**: Operational outcomes reflect that tasks were completed (e.g., "Activity moved indoors"). Measured impact requires sensor evidence showing conditions improved.
- **Forecast validation and historical ground truth**: The comparison of past forecasts against actual measurements to determine model error (MAE/RMSE).

Section 4 — Getting Started

AIRWISE runs locally in your web browser. When you start the application, you will arrive at the **Command Center** dashboard.

- **Navigation**: The left sidebar provides access to all major sections: Command Center, Air Intelligence, Campus Assessment, Response Centre, History, and Forecast Lab.
- **Campus Selection**: At the top right of the application, you can select your active campus (e.g., Ghaziabad Campus). Changing this updates all data context.
- **Returning to Dashboard**: Click the AIRWISE logo or the "Command Center" link in the sidebar.

Section 5 — Page-by-Page Guide

1. Command Center

- **Purpose**: A high-level overview of active plans, recent assessments, and campus status.
- **Information**: Displays active draft and approved plans, recent incidents, and quick status tiles.
- **Action**: Use this to quickly navigate to pending approvals or ongoing tasks.

2. Air Intelligence

- **Purpose**: Displays raw environmental data and forecasts.
- **Information**: Current pollutant levels, 24-hour forecast timelines, and source metadata.
- **Action**: Review this page to understand raw pollutant trends before generating an assessment.

3. Campus Assessment

- **Purpose**: Evaluates current conditions against an activity type to generate risk recommendations.
- **Information**: Activity dropdown, assessment result, risk thresholds, and Data Quality Panel.
- **Action**: Select an activity (e.g., Outdoor Sports), review the assessment, and use the Decision Simulator to draft a Response Plan.
- **Empty/Error State**: If data fails to load, an error banner appears. You must wait for network restoration.

4. Response Centre

- **Purpose**: Manages the lifecycle of Action Plans.
- **Information**: Lists Draft, Approved, and Completed plans. Displays tasks and decision receipts.
- **Action**: Administrators can approve/dismiss draft plans. Task owners update task statuses. Once completed, outcomes are recorded here.

5. History / Reports

- **Purpose**: The permanent archive of all completed operations and incidents.
- **Information**: Filterable lists of past assessments, outcomes, and logged incidents.
- **Action**: Export the ledger to JSON for compliance audits using the Export button.

6. Forecast Lab

- **Purpose**: Evaluates the historical accuracy of the forecast model.
- **Information**: Explains Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).
- **Empty State**: If historical ground-truth data from CPCB is missing, the page honestly explains that validation cannot be calculated.

Section 6 — Complete Operational Walkthrough

1. **Select a campus**: Use the top-right dropdown to select "Ghaziabad Campus".

2. **Review data**: Navigate to Air Intelligence to view current modelled forecasts.
3. **Generate assessment**: Go to Campus Assessment, select an activity, and click "Assess Current Forecast".
4. **Understand result**: Review the output (e.g., "Review Required"). Note the exact PM2.5 levels shown in the evidence block.
5. **Review alternatives**: Scroll down to the Decision Simulator.
6. **Compare scenarios**: Review the impact of "Proceed", "Move Indoors", or "Reschedule".
7. **Select alternative**: Click "Draft Plan" on your chosen alternative (e.g., Move Indoors).
8. **Create response plan**: Confirm the modal. A plan is generated.
9. **Review decision receipt**: Go to Response Centre, find your Draft plan, and click "View Decision Receipt".
10. **Verify receipt**: Click "Verify Integrity". The system hashes the payload and confirms it hasn't been tampered with.
11. **Submit for approval**: Click "Approve Plan".
12. **Understand approval**: The plan moves to "Approved" state. Tasks are now unlocked.
13. **Review tasks**: View assigned tasks (e.g., "Notify participants").
14. **Progress tasks**: Mark tasks as "Start" and then "Complete". The plan auto-completes when all tasks finish.
15. **Record outcome**: Click "Record Outcome" on the completed plan.
16. **Distinguish outcome**: Choose "Operational Outcome Only" if you don't have sensor data proving air quality improved.
17. **Find in History**: Go to History to see the archived plan and its recorded outcome.
18. **Export**: Click "Export Audit JSON" to download the records.

Disclaimer: Scenarios are decision support only, not medical advice.

Section 7 — Understanding Air-Quality Information

AIRWISE displays both modelled forecasts (Open-Meteo) and optional ground observations.

- **Modelled Forecasts**: These represent expected conditions across a regional grid. They are NOT point-specific ground measurements.
- **Averaging Periods**: Modelled data is often provided hourly. Hourly spikes do not equate to a 24-hour regulatory average violation. Thresholds shown in AIRWISE apply specifically to the units and periods labelled.
- **Data Gaps**: Stale or missing data will trigger warnings in the Data Quality Panel. A suggested forecast time window is a computational estimate, not a guarantee of safe outdoor conditions.

Section 8 — Data Readiness Score and Evidence

The Data Readiness Score evaluates the freshness and completeness of the ingested environmental data.

- It is based entirely on the presence of recent timestamps and the absence of network dropouts.
- It is **not** a calibrated probability of forecast accuracy.
- If the score is low or warnings appear, the assessment relies on degraded evidence, and users should exercise heightened caution.

Section 9 — Decision Receipts and Verification

When a plan is created, AIRWISE generates a **Decision Receipt**.

- **Purpose**: To bind the chosen operational plan to the exact environmental evidence present at that exact moment.

- **Verification**: The system uses a deterministic SHA-256 hash. If someone manually alters the database to fake a lower PM2.5 reading, the hash verification will fail.
- **Limitations**: The receipt is tamper-evident, not strictly immutable. It does not use cryptographic digital signatures tied to external trust authorities (like a blockchain or PKI).

Section 10 — Response Plans, Approvals, and Tasks

- **Lifecycle**: Draft → Approved → Completed (or Dismissed).
- **Roles**: Campus Administrators approve plans. Coordinators execute tasks. (Roles are enforced by operational policy in the demo, not authenticated sessions).
- **Failures**: If an action fails, an error banner appears, and the action state remains unchanged.

Section 11 — Outcomes, History, and Exports

- **Measured vs Operational**: Completing a task (e.g., turning on purifiers) is an operational outcome. It does not prove the air got cleaner. You must select "Measured Impact" only if you have physical sensor evidence. Otherwise, use "Operational Outcome Only".
- **Exports**: History can be exported as a JSON blob.
- **Persistence Limitations**: Data is stored in memory for the local demo. Restarting the backend server deletes all history.

Section 12 — Forecast Lab and Model Validation

The Forecast Lab exists to compare past forecasts against real ground truth. Because historical sensor data is not continuously ingested in the default configuration, the Lab will typically display an "Insufficient Data" state. AIRWISE intentionally does not fabricate accuracy metrics.

Section 13 — Troubleshooting and FAQ

- **Website fails to load**: Ensure `npm run dev` is running on port 3000.
- **Backend/API Unavailable**: Ensure `node local.mjs` is running on port 3001. If it was closed, your data has been reset.
- **Missing Forecast Data**: Check your internet connection; Open-Meteo requires outbound network access.
- **Receipt Verification Failing**: The underlying assessment data was modified. The audit trail is compromised.

Section 14 — Privacy, Data Integrity, and Limitations

- **Data Sources**: Relies on public APIs (Open-Meteo).
- **Persistence**: Local deployment uses temporary in-memory storage. AWS deployment (when configured) uses DynamoDB.
- **Authentication**: The current hackathon demo uses trust-based role selection without enforced password authentication.
- **Privacy**: No Personally Identifiable Information (PII) is required to operate AIRWISE.